

Francesco Sammarco

Data Scientist & AI Engineer Consultant, AI Researcher

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PROFILE

AI/ML Engineer with approximately 5 years of experience building customer-facing software, ML-based solutions, cloud infrastructure, delivery automation and, more recently, agentic platforms. Currently designs and ships enterprise multi-agent, knowledge-management, and agentic analytics systems, while dedicating to research in DL-based architectures for medical imaging.

WORK EXPERIENCE

AI/ML Engineer Consultant

Mar. 2026 – Present

Live Reply

Remote / Turin, IT

- Delivered an enterprise GenAI knowledge-management system over 200+ technical documents, reducing operator search time by 52% through ingestion pipelines, retrieval evaluation, Qdrant, Neo4j, and agentic GraphRAG workflows.
- Designed an Azure-based agentic analytics platform that converts transactional data into validated, actionable reports, reducing recurring report preparation time by 81% through SQL and object-storage integration, specialized agents, workflow memory, and automated quality checks.
- Built a stateful multi-agent orchestration platform with Python and LangGraph, enabling non-technical users to automate several business processes.
- Provisioned reproducible AWS environments with Terraform, Kubernetes, networking, load balancing, AgentCore runtime, API gateways, and containerized AI services, supporting separate development, testing, and production configurations.
- Led GenAI coding-platform adoption across 4 engineering teams by defining agent workflows, repository-aware development practices, and reusable tool integrations for multiple software-delivery use cases.

Applied AI Researcher

Sep. 2024 – Present

University of Tübingen

Remote / Tübingen, DE

- Built a reproducible PyTorch and SLURM pipeline for distributed medical-image training, inference, experiment tracking, and explainability evaluation across approximately 70,000 DICOM/NIFTI images and 300,000 clinical records.
- Developed generative-AI and explainable-AI methods for psychiatric neuroimaging, combining diffusion-based counterfactual generation, computer vision, statistical evaluation, and clinically oriented model interpretation.
- Contributed to two open-source medical-AI projects covering foundation-model merging and neuroimaging explainability: <https://huggingface.co/guinansu/MedSAMix> and my paper which is currently being finalized.

Applied AI Researcher, Short-Term Collaboration

February 2026 – March 2026

BeyondShape

Naples, Italy

- Delivered 2 applied-AI pipelines: a 3D spine-reconstruction workflow using gradient boosting, convolutional networks, and generative methods, and an isolation-forest anomaly detector for large-scale ISIC dermatology data.

DevOps Engineer, Telecommunications Consultant

2022 – 2024

Hewlett Packard Enterprise

Milan, Italy

- Built a Jenkins, Java, Spring, and SQL deployment system for 1 major telecommunications customer, reducing release execution time by approximately 90% while improving repeatability and operational reliability.
- Owned delivery across 3 stakeholder groups (client representatives, software developers, and operations engineers), translating business requirements into technical designs, implementation plans, and production releases.
- Implemented Java and SQL backend integrations and introduced containerized delivery components, removing production blockers across multiple release cycles.

IT Consultant

2020 – 2021

Concept Solution

Pompeii, Italy

- Supported 4+ digital-transformation initiatives spanning robotic process automation, customer profiling, blockchain traceability, and e-commerce recommendation systems by translating business requirements into technical solution designs.

EDUCATION

Politecnico di Milano

Sep. 2020 – Oct. 2022

Master of Computer Science and Engineering (ENG) | 110 cum laude/110, GPA: 29.43/30

Milan, IT

- Artificial Intelligence focused path, thesis on Reinforcement Learning.

Università degli Studi di Napoli Federico II

Sep. 2017 – Jul. 2020

Bachelor of Computer Science and Engineering (IT) | 110 cum laude/110, GPA: 29.70/30

Naples, IT

- Background in computer science, mathematics, and physics (including databases, software engineering, and cybersecurity).

SELECTED ACHIEVEMENTS

- ACM RecSys Challenge 2022** Mar. 2022 – Jul 2022
- Ranked 29th out of 250+ teams in the Dressipi session-based recommendation challenge, contributing ranking models and seasonality features later published in the challenge proceedings.
- AI Summer School Instructor** Sep. 2025
- Designed and delivered hands-on AI teaching sessions for health and life-science participants, translating neuroimaging ML concepts into practical exercises with reproducible [code](#) [data](#).
- Apple Academy** Sep. 2019 – Aug. 2020
- Ranked among top 100 and accessed a one year internship focused on app development in Swift and SwiftUI and project management fundamentals.
- Cyber Challenge** May 2018 – Jul. 2018
- Ranked among top 50 and accessed closed number course focused on cybersecurity fundamentals, with execution of challenges on networking, reverse engineering, pwning and cryptography.

PROJECTS

- Thesis: Non-Stationary Reinforcement Learning for Trading** Sep. 2021 – Oct. 2022
Title: Non-Stationary RL for Forex Trading with Automatic Market Regime Clustering
- Developed a machine learning-based automatic trading system compatible with Forex assets, exploiting [FQI](#) reinforcement learning algorithm (with [gym](#)) and a mixture of clustering and gradient-boost based classification ([xgboost](#)) techniques for timeseries data.
- Recommender System Challenges (RecSys 2022, H&M, Internal)** Mar. 2022 – Jul 2022
- Developed recommender systems across media and fashion domains using collaborative filtering, matrix factorization, *SLIM*, *GRU4Rec*, and LightGBM-based ranking; contributed to ACM RecSys 2022 Dressipi challenge solution ranking 29th / 250+ teams with 0.1845 MRR private leaderboard score and related publication. [Github Repo](#) (paper in [Publications](#)). Trained using AWS EC2 spot instances.
- Artificial Neural Network Challenges** Oct 2020 – Jan 2021
- Solved multiple applied AI challenges across image detection/segmentation, text classification, and sequence modeling, designing neural-network pipelines with YOLO/RNN-based architectures and benchmarking performance on competitive leaderboards; 2/3 ended in top five. Results available at [Kaggle profile](#).
- Thesis on Genetic Algorithms** Apr. 2020 – Jul. 2020
Title: Application of Genetic Algorithms on Flexible Job Shop Scheduling problem
- Designed and implemented a Python metaheuristic optimizer for Flexible Job Shop Scheduling, improving production planning decisions and times by jointly optimizing machine assignment and job sequencing; code available at [GitHub repo](#).

PUBLICATIONS

- Yang Y.*, Su G.*, Hu J.*, Sammarco F., Geiping J., Wolfers T.,** *Medsamix: A training-free model merging approach for medical image segmentation.*, arXiv preprint arXiv:2508.11032 (*AAAI*), (2025) - [DOI](#)
- Della Volpe N.*, Mainetti L.*, Martignetti A.*, Menta A.*, Pala R.*, Polvanesi G.*, Sammarco F.*, Maurera F. B. P., Bernardis C., and Dacrema M. F.,** *Lightweight Model for Session-Based Recommender Systems with Seasonality Information in the Fashion Domain.*, In Proceedings of the Recommender Systems Challenge, (2022) - [DOI](#)

TECHNICAL SKILLS

Programming Languages: Python, Cython, Java, SQL, C/C++, Swift, C#, JavaScript, MATLAB, HTML/CSS, Kubernetes
ML/Data Libraries: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, OpenCV, nilearn, matplotlib, gym, langgraph, langchain, litellm, pydantic, crewai
Frameworks & Tools: Spring, JPA/Hibernate, Docker, Kubernetes, Jenkins, Git, GitHub/GitLab, AWS, Ansible, Vagrant, Wandb, MLFlow, Qdrant, Neo4j

Languages: *Italian:* C2 (native), *English:* C1, *German:* A2 (studying)